

1 1D ODE Problem

Definition 1.1. The 1 dimensional ODE problem is posed as such, given,

$$x'(t) = f(x(t)), \quad x(0) = x_0$$

we want to find a curve $x : I \rightarrow U$ which has the property that $x'(t) = f(x(t))$ and $x(0) = x_0$.

Example 1.2. Consider $f(x) = x$, with $x'(t) = x$, then $x(t) = xe^t$.

Definition 1.3. Let $X : U \rightarrow \mathbb{R}^n$, $U \subseteq \mathbb{R}^n$ be a vector field (assume X is C^1). A curve $\gamma : I \rightarrow U$ is said to be an integral curve of X if

$$\gamma'(t) = X(\gamma(t)).$$

Writing

$$\gamma(t) = \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix}, \quad \gamma'(t) = \begin{pmatrix} x'_1(t) \\ \vdots \\ x'_n(t) \end{pmatrix}, \quad X(\gamma(t)) = \begin{pmatrix} X_1(\gamma(t)) \\ \vdots \\ X_n(\gamma(t)) \end{pmatrix},$$

we have that

$$\frac{dx_i}{dt} = X_i(\gamma(t)), \quad 1 \leq i \leq n.$$

Furthermore, if $\gamma(0) = \gamma_0$ then we say that γ is an integral curve *originating from* γ_0 .

Example 1.4. Consider the function,

$$X(x, y) = (-y, x).$$

Then we have according to the ODE form, $x' = -y$ and $y' = x$. We can simplify as,

$$x'' = (-y)' = -y' = -x \implies x'' + x = 0$$

This is simply the SHM equation with the solution being,

$$x(t) = A \sin t + B \cos t.$$

Given $(x_0, y_0) = (x(0), y(0))$, we have $x(0) = B = x_0$. Also,

$$x'(t) = A \cos t - B \sin t,$$

and since $y = -x'$, $x(0) = B = x_0$. Thus

$$x(t) = y_0 \sin t + x_0 \cos t, \quad y(t) = -y_0 \cos t + x_0 \sin t$$

Observe that the following quantity is a constant (or *conserved*)

$$x(t)^2 + y(t)^2 = x_0^2 + y_0^2,$$

so the equation $x^2 + y^2 = \text{constant}$ describes the circles traced by the integral curves.

Theorem 1.5 (Picard theorem). We have $F : I \times U \rightarrow \mathbb{R}^n$, where I is an interval with non-empty interior and $U \subseteq \mathbb{R}^n$ is open. Assume, F is continuous and fixing $(t_0, y_0) \in I \times U$. We also assume that there exists $W \subseteq I \times U$ open neighbourhood of (t_0, y_0) such that, for some $L > 0$, $\forall (t, y), (t, y_1) \in W$,

$$\|F(t, y) - F(t, y_1)\| \leq L \|y - y_1\|$$

i.e. locally lipschitz in the y variable. Then there exists $\epsilon > 0$ such that in the open interval $(t_0 - \epsilon, t_0 + \epsilon)$ there exists unique solution to the equation $y(t)$ such that $y'(t) = F(t, y(t))$ and $y(t_0) = y_0$.

Exercise 1.6. If f is C^1 , show f is *locally Lipschitz*. (use MVT)

Proof.

$$y(t) = y(t_0) + \int_{t_0}^t y(s) ds$$

$$\underline{y}(t) = y(t_0) + \int_{t_0}^t F(s, \underline{y}(s)) ds$$

Observe that y is input on RHS and output on LHS. From an operator's Point of View, this motivates looking at it like a *Fixed Point Problem* as follows,

$$(Tu)(t) = y_0 + \int_{t_0}^t F(s, u(s)) ds.$$

Note. y is a solution to the ODE iff y satisfies the integral equation

$$y(t) = y_0 + \int_{t_0}^t F(s, y(s)) ds.$$

Further, if a continuous y has this property, then by the FTC,

$$y'(t) = F(t, y(t)), \quad y(t_0) = y_0.$$

Thus $y(t)$ is a solution to the ODE problem.

Since F is defined on $I \times U$ and is Lipschitz on an open set W , we can find $b, r > 0$ such that

$$[t_0 - b, t_0 + b] \times B_r(y_0) \subseteq W.$$

Also, let

$$M = \sup\{\|F(t, y)\| : (t, y) \in V\}.$$

We choose $\epsilon > 0$ such that

(The third condition will be needed to show that the map T is a self-map.)

Note. X is a complete metric space when equipped with the uniform norm

$$\|u\|_\infty = \sup_{t \in J} \|u(t)\|.$$

Indeed, this norm induces a metric on $C(J, \mathbb{R}^n)$, and X is closed in $C(J, \mathbb{R}^n)$. Hence X is complete.

Now on X , define the operator $T : X \rightarrow C(J, \mathbb{R}^n)$ by $u \mapsto Tu$, where

$$(Tu)(t) = y_0 + \int_{t_0}^t F(s, u(s)) \, ds.$$

We claim that T maps X into X . To prove we fix $u \in X$,

$$\begin{aligned} \|(Tu)(t) - y_0\| &= \left\| \int_{t_0}^t F(s, u(s)) \, ds \right\| \\ &\leq \int_{t_0}^t \|F(s, u(s))\| \, ds \\ &\leq M |t - t_0| \\ &\leq M\epsilon \leq r. \end{aligned}$$

Hence $Tu \in X$, which proves the claim. Second, we claim that T is a contraction. To prove, we fix, $u, v \in X$

$$\begin{aligned} \|(Tu)(t) - (Tv)(t)\| &= \left\| \int_{t_0}^t (F(s, u(s)) - F(s, v(s))) \, ds \right\| \\ &\leq \int_{t_0}^t \|F(s, u(s)) - F(s, v(s))\| \, ds \\ &\leq L \int_{t_0}^t \|u(s) - v(s)\| \, ds \\ &\leq L |t - t_0| \|u - v\|_\infty \\ &\leq L\epsilon \|u - v\|_\infty \\ &\leq \|u - v\|_\infty. \end{aligned}$$

Taking supremum over $t \in J$,

$$\|Tu - Tv\|_\infty \leq \|u - v\|_\infty.$$

By Banach's Contraction Principle, there exists a unique $y \in X$ such that $Ty = y$. Hence y solves the ODE.

Definition 1.7 (General ODE problem). Consider $X : \mathbb{R}^n \rightarrow \mathbb{R}^n$ a vector field, and the problem

$$\gamma'(t) = X(\gamma(t)), \quad \gamma(0) = y_0.$$

Typically, γ is C^1 . Writing $F(t, y) = X(y)$ if X is C^1 , then F is globally continuous and *locally Lipschitz* in y .

Theorem 1.8 (Picard–Lindelöf). We can locally find an integral curve for the vector field X originating at x_0 . We can then define our flow function

$$\Phi_t(x) = \gamma_x(t),$$

at least locally around $t = 0$, where γ_x is the integral curve originating at x .

Example 1.9. Let $X(x) = x$ then $\Phi_t(x) = xe^t$.

Remark 1.10. If we fix x and vary t , we obtain the integral curve through x . On the other hand, if we fix t and vary x , then $\Phi_t(x)$ gives the position at time t of the integral curve which started at x .

Immediately from the above expression we have,

$$\Phi_0(x) = x \implies \Phi_0 = \text{id}. \quad (1)$$

$$\Phi_{t+s}(x) = \Phi_t(\Phi_s(x)) = \Phi_s(\Phi_t(x)) \quad (2)$$

$$\Phi_{-t} \circ \Phi_t = \text{id}, \quad (\Phi_t)^{-1} = \Phi_{-t}. \quad (3)$$

The second expression follows from uniqueness of the integral curves. The third expression indicates that $t \mapsto \Phi_t$ is a homomorphism (roughly) to a family of functions.

Now suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Then

$$\frac{d}{dt}f(\Phi_t(x)) = Df(\Phi_t(x)) \frac{d}{dt}\Phi_t(x).$$

Since

$$\frac{d}{dt}\Phi_t(x) = \frac{d}{dt}\gamma_x(t) = X(\gamma_x(t)) = X(\Phi_t(x)),$$

Substituting we get

$$\frac{d}{dt}f(\Phi_t(x)) = Df(\Phi_t(x))X(\Phi_t(x)).$$

Writing $X = (X_1, \dots, X_n)$ and

$$Df(x) = \left(\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_n}(x) \right),$$

we have

$$\frac{d}{dt}f(\Phi_t(x)) = \sum_{i=1}^n X_i(\Phi_t(x)) \frac{\partial f}{\partial x_i}(\Phi_t(x)).$$

Remark 1.11. If

$$\sum_{i=1}^n X_i(x) \frac{\partial f}{\partial x_i}(x) = 0 \quad \forall x \in U,$$

then

$$\frac{d}{dt}f(\Phi_t(x)) = 0.$$

Hence f is conserved along the integral curves of X .

□